

ONKAR ANIL PEDNEKAR

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SUMMARY

Software Engineer with around 2 years of experience in backend development, web applications, and machine learning. Skilled in Python, Flask, Django, Java, and SQL, with expertise in building scalable systems, database management, and automation tools. A proactive problem-solver and team player with a solid foundation in system design and data structures

WORK EXPERIENCE

Software Engineer – Meta (Contract) – Austin, TX

Feb 2024 – Present

- Optimized a **PyTorch-based deep learning model** for AutoDocs, improving support question summarization by 25% through data preprocessing techniques including lemmatization, tokenization, and TF-IDF vectorization. **and fine-tuning on domain-specific data**, achieving 90% accuracy in question categorization.
- Integrated a Bogus Response Rate health check using C++ for analyzing ML models with high NaN/Inf responses on production traffic, and visualizing the failure with a timeseries chart in the UI **reducing model errors by 40%**.
- Developed a Python-based automation tool to diagnose production ML deployment issues by querying databases, parsing logs, and analyzing pipeline failures via API calls, reducing **issue resolution time by 40%**.
- Implemented Python-C++ integration using Apache Thrift to block ML model publishing to a legacy platform, enforcing migration to the new system and **reducing incorrect deployments by 95%**.
- Mitigated ML model deployment failures using internal tools, enhancing system scalability and reliability by 30%.

Machine Learning Research Assistant – University of North Carolina At Charlotte – Charlotte, NC

May 2023 – Aug 2023

- Collaborated with a Machine Learning Engineer to develop a **convolutional neural network model with LSTM layers for 3D pose estimation**, using 2D pose coordinates from MMPose to generate accurate 3D coordinates.
- Developed an image generation model using **Stable Diffusion SDXL** for mesh reconstruction to generate a human mesh from 3D coordinates, for sports analysis, predicting player performance and injury risk with 95% efficiency.

Freelance Machine Learning and Software Engineer – Upwork Inc – Remote

Apr 2021 – Jan 2022

- Engineered 45+ AI/ML solutions using PyTorch and TensorFlow, including an YOLOv5 **Object Detection** model for tracking golf club heads, **Face Recognition**-based attendance system, and a fine-tuned Llama model for **Text Summarization**, all delivered on time and within budget, achieving **97% customer satisfaction**.

Machine Learning and Data Science Intern – Cyberace Infovision Pvt. Ltd. – Mumbai, India

Feb 2021 – Mar 2021

- Leveraged **Logistic Regression** to forecast product sales correlations with discounts, leading to a **15% revenue increase** through data-driven pricing strategies, while employing **GridSearchCV for hyperparameter tuning**.
- Conducted data analysis on 10,000+ data points using **K-Means Clustering** and Seaborn to extract actionable insights.

TECHNICAL SKILLS

- Programming Languages:** Python, Java, C, C++, JavaScript, HTML, SQL
- Databases:** Relational (MySQL, MS SQL Server, SQLite, AWS RDS), NoSQL (AWS DynamoDB, PostgreSQL, MongoDB)
- Tools/Technologies:** Java Swing, Java Spring, Rest APIs Flask, Django, Hadoop, Spark, Scikit-learn, PyTorch, Tensorflow
- Platform:** AWS (S3, EC2, Lambda, Sagemaker, Tableau, Git, Anaconda, Eclipse, VS Code, Azure)
- Data Science:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Generative AI, Statistics

EDUCATION

Master of Science in Computer Science

Jan 2022 – Dec 2023

University of North Carolina at Charlotte – Charlotte, NC

GPA – 3.81 / 4.0

Bachelor of Engineering, Computer Engineering

Aug 2017 – May 2021

University of Mumbai – Mumbai, India

GPA – 3.65 / 4.0

ACADEMIC PROJECTS

Virtual Wardrobe: Text-to-Image Synthesis with StackGAN and PyTorch

Dec 2023

- Led the development of **text-to-image** application which takes clothing description as input and generates images of people wearing those clothes as output, leveraging **StackGAN** architecture consisting of two **adversarial networks**.

TODO Web Application using Flask, HTML and CSS

May 2023

- Engineered an innovative application using **Flask** for **backend**, with **HTML and CSS** for **front-end** Web Development, enabling users to seamlessly add, edit, and delete tasks, resulting in a 20% increase in task management efficiency.
- Integrated **SQLITE database** using **SQLAlchemy**, achieving a **25% improvement in data storage and retrieval operations**, and refined the user interface for seamless task manipulation and user experience.

Travel Agency using Java

Jan 2023

- Developed a Java-based flight and hotel booking system with **MySQL database** connectivity, utilizing object-oriented programming (OOP) techniques, and **Java Swings** for **UI design**, for an efficient user interface.